

CODEHIVE

Level up your Java: Numbers & Strings Guide

SYSTEM ARCHITECTURE: The + operator is overloaded. Result depends on type precedence.

Scenario 1: Numeric Addition

```
int x = 10;
int y = 20;
int z = x + y; // Result: 30
System.out.println(z);
```

Scenario 2: String Concatenation

```
String x = "10";
String y = "20";
String z = x + y; // Result: "1020"
System.out.println(z);
```

Scenario 3: Mixed Types

```
String x = "10";
int y = 20;
```

```
String z = x + y; // Result: "1020"  
System.out.println(z);
```

Core Takeaways

- ✓ **Numerical:** Addition happens only when both sides are numbers.
- ✓ **Literal:** Concatenation takes over if either side is a String.
- ✓ **Coercion:** Values are automatically converted to strings in mixed operations.
- ✓ **Precedence:** Use () to force math before text joining.